ASHUTOSH SONAR

DEVOPS ENGINEER | CLOUD PLATFORM ENGINEER

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PROFESSIONAL SUMMARY

DevOps Engineer with **4+ years of experience** designing and scaling **AWS-native infrastructure**, CI/CD pipelines, and secure cloud platforms in regulated environments. Proven track record of improving deployment speed, system reliability, and cloud security posture using Infrastructure as Code, Kubernetes, and observability tooling. Experienced in building **HIPAA- compliant, high-availability systems**, enabling large-scale data and ML workloads. Strong expertise in **cloud security, automation, and SRE practices**, with a focus on performance optimization and zero-downtime delivery.

TECHNICAL SKILLS

Programming & Scripting: Python, Bash, Shell, Go
Cloud Platforms: AWS (EC2, S3, Lambda, IAM, KMS, VPC, EKS, CloudWatch), Databricks
DevOps & CI/CD: Jenkins, ArgoCD, GitHub Actions, GitOps
Containerization & Orchestration: Docker, Kubernetes(EKS)
Infrastructure as Code: Terraform, Ansible
Monitoring & Observability: Prometheus, Grafana, ELK Stack, CloudWatch
Databases & Messaging: Neo4j,PostgreSQL, MongoDB, Redis, Apache Kafka
Security & Compliance: IAM, KMS, Secrets Management, HIPAA Compliance, Threat Modeling (STRIDE, PASTA, LINDDUN, DREAD)
Practices: SRE, IaC, Microservices, Zero-Downtime Deployment, Agile/Scrum, Observability Engineering

PROFESSIONAL EXPERIENCE

Florida Institute of Technology
Cloud Platform Engineer | Jan 2026 – Present

- Architected and deployed a petabyte-scale, multi-tenant healthcare data platform on AWS and Databricks, enabling secure access for 200+ researchers and reducing data delivery time from days to same-day.
- Designed and enforced HIPAA-compliant cloud architecture, implementing IAM least-privilege access, KMS encryption, VPC segmentation, and centralized audit logging, ensuring full regulatory compliance.
- Automated infrastructure provisioning using Terraform and Python, reducing environment setup time by 85% and enabling consistent deployments across dev, staging, and production.
- Implemented end-to-end observability (CloudWatch + ELK Stack), improving anomaly detection time by 70% and reducing incident response time by 30%.
- Optimized Databricks infrastructure to support 15–20 concurrent ML workloads, improving compute efficiency and reducing onboarding time for new teams by 60%.
- Collaborated with cross-functional teams to align platform capabilities with data science and clinical workflows, improving system usability and adoption.
- Strengthened platform reliability through proactive monitoring, alerting, and incident management aligned with SRE principles.

Graduate Research Assistant | Jul 2025 – Dec 2025

- Built containerized ML infrastructure using Docker and Kubernetes(EKS),reducing environmentsetup time by 80% and enabling self-service capabilities for research teams.

- Integrated automated threat modeling (STRIDE, PASTA, LINDDUN, DREAD) into CI/CD pipelines, preventing critical misconfigurations before production deployment.
- Enhanced pipeline security by embedding policy checks, validation gates, and compliance controls, improving deployment quality and reducing risk exposure.
- Analyzed system telemetry and logs to identify recurring reliability gaps, increasing early detection rates by 40% across multiple deployment cycles.

Coriolis Technology

DevOps Engineer (Member of Technical Staff) | Oct 2020 – Jul 2023

- Designed andmanaged CI/CD pipelines(Jenkins, ArgoCD) for 20+ microservices,reducing release cycles by 50% and increasing deployment frequency.
- Built and maintained EKS-based Kubernetesinfrastructure, ensuring 99.999% uptime with zero-downtime deployment strategies and rolling updates.
- Automated cloud provisioning and deployment workflows using Python and Bash, cutting manual effort by 80% and improving release consistency.
- Implemented Infrastructure as Code (Terraform, Ansible) acrossmulti-account AWS environments, eliminating configuration drift and improving scalability.
- Strengthened cloud security posture by integrating IAM policies, KMS encryption, and secrets management, reducing deployment-related security failures by 65%.
- Developed comprehensive monitoring solutions using Prometheus, Grafana, and CloudWatch, reducing MTTR by 40% and improving incident resolution speed.
- Improved system resilience through proactive alerting, logging, and performance tuning across distributed microservices architecture.
- Collaborated with engineering teams to implement DevOps best practices, improving development velocity and deployment reliability.

PROJECTS

TF-Lens

- Developed an open-source Go CLI tool that converts Terraform files into interactive, hierarchical infrastructure diagrams, improving on 'terraform graph' with zero external dependencies.
- Created a diff mode with color-coded indicators to visualize infrastructure drift during terminal-based PR reviews.

YAMLGuard – CI/CD Configuration Validator

- Built a CI/CD validation tool to enforce **YAML schema compliance and configuration integrity**,reducing deployment failures and improving pipeline reliability.

EDUCATION

Florida Institute of Technology

Master of Science in Computer Science | Aug 2023 – May 2025

- Thesis: Integrating Threat Modeling (STRIDE, PASTA, DREAD) into CI/CD Pipelines

Savitribai Phule Pune University

Bachelor of Engineering in Computer Engineering | Aug 2016 – May 2020